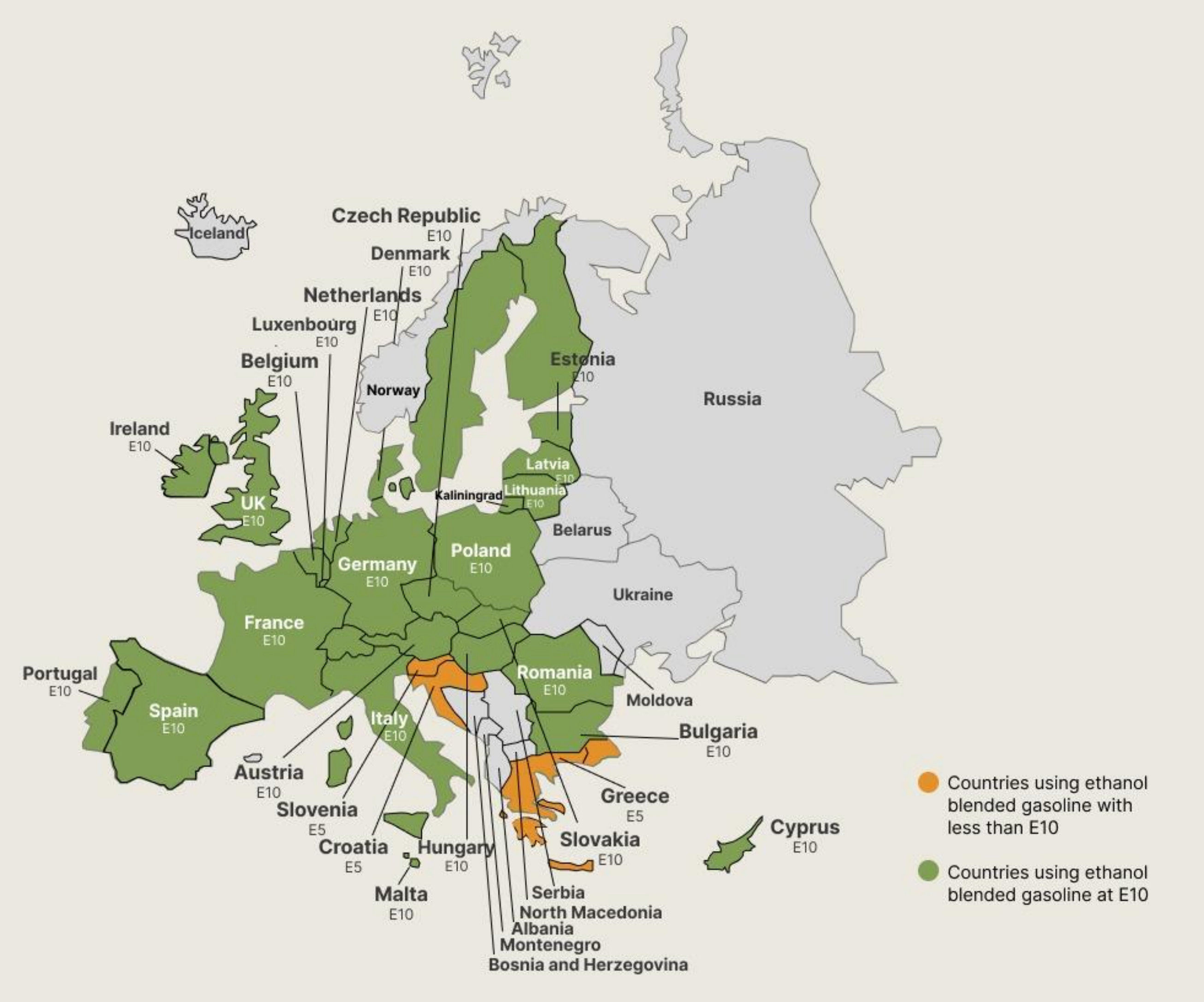
Ethanol Blending in Gasoline - Poland

September, 2025

Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challanges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

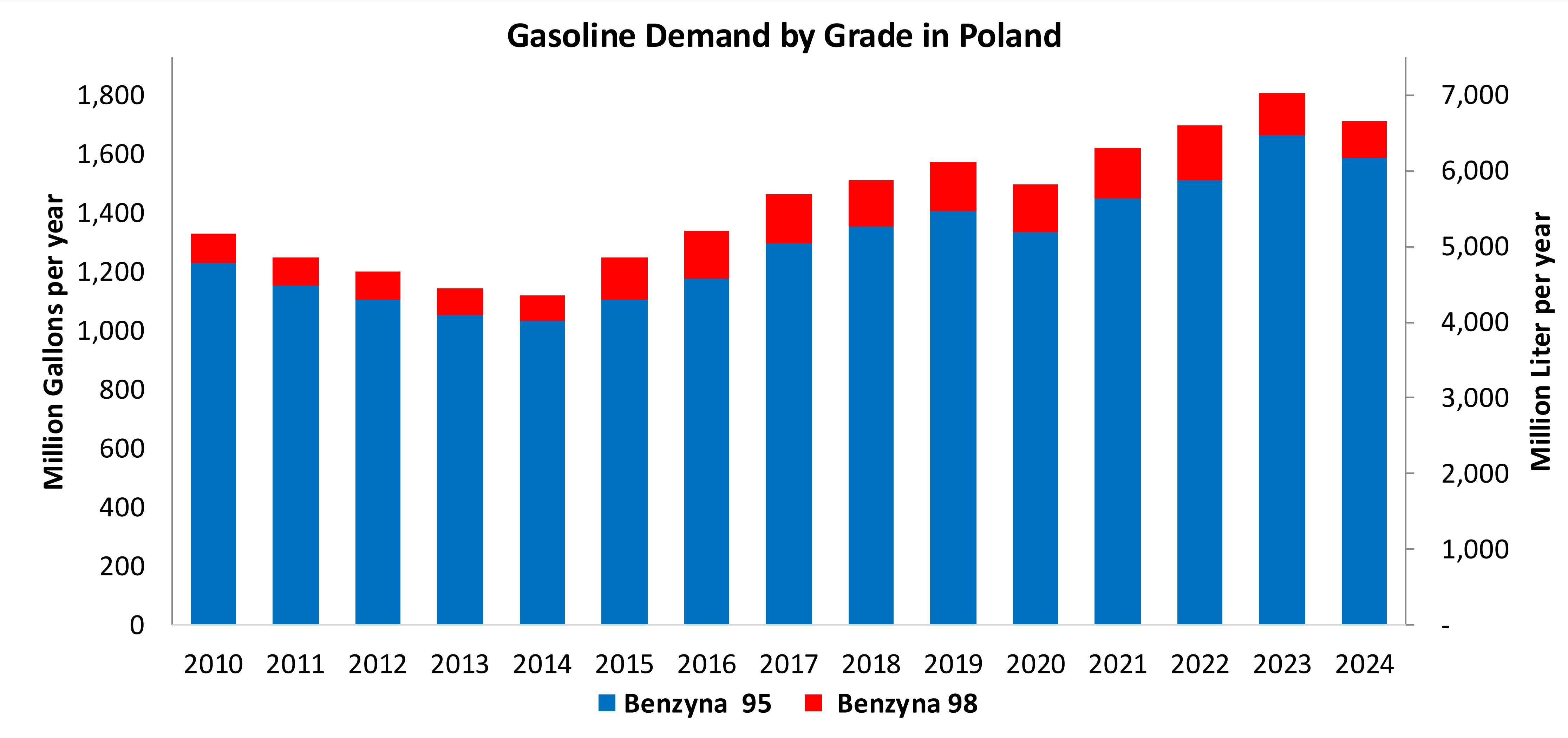
Ethanol Gasoline Blending in Poland

In 2024, gasoline consumption in Poland was 6,493 million liters (1,715 million gallons) and growth rate was 1.7%. Gasoline production and net imports were 5,606 and 913 million liters (1,481 and 241 million gallons) correspondingly. Poland complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.2 RON.

In 2024, ethanol consumption in Poland was 420 million liters (111 million gallons) representing 6.5% of gasoline demand. Ethanol production and Net Imports were 452 and -32 million liters (111 and -9 million gallons) correspondingly. Ethanol sources are Corn 57% and Cereals 43%.

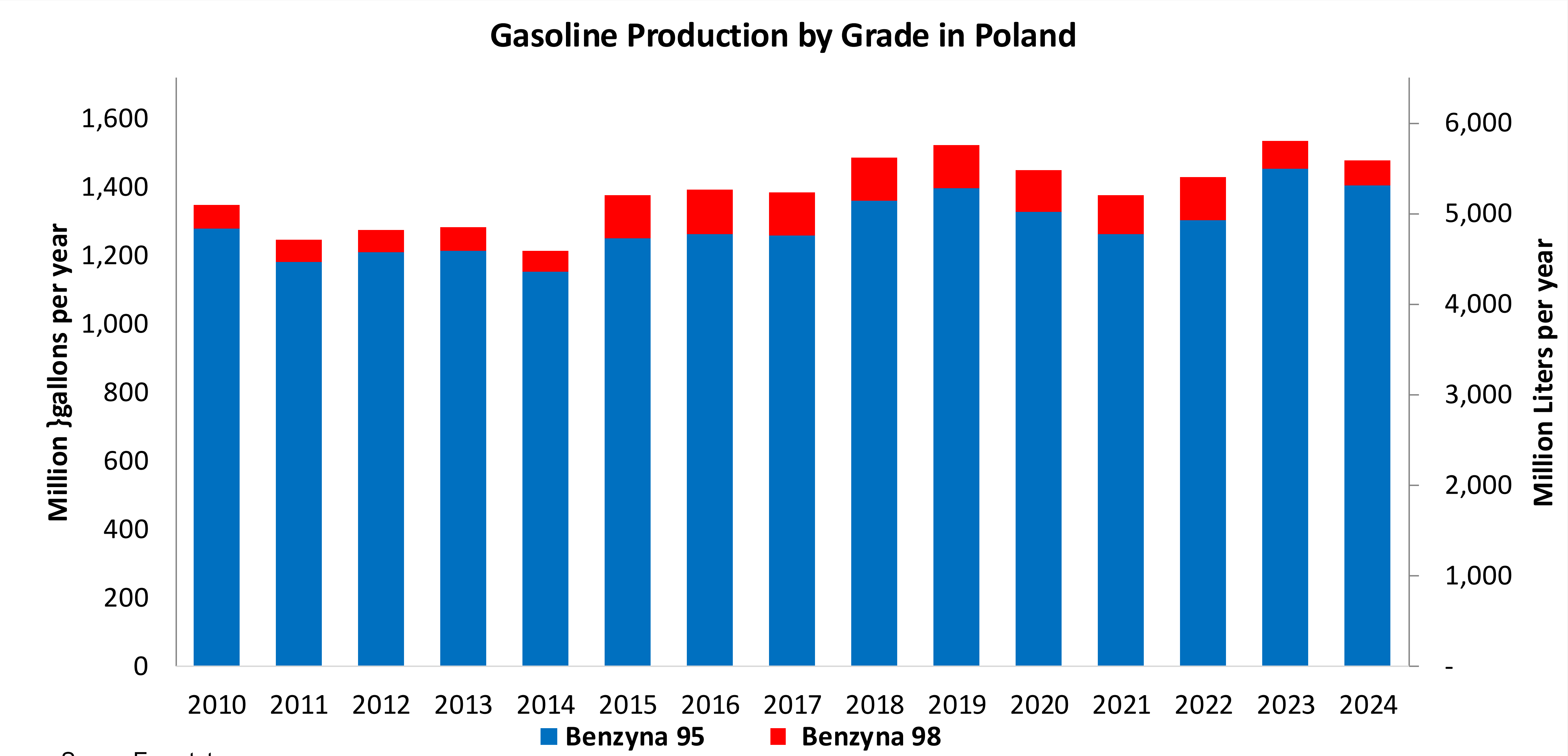
Source: Eurostat, European Commision

Gasoline Demand in Poland



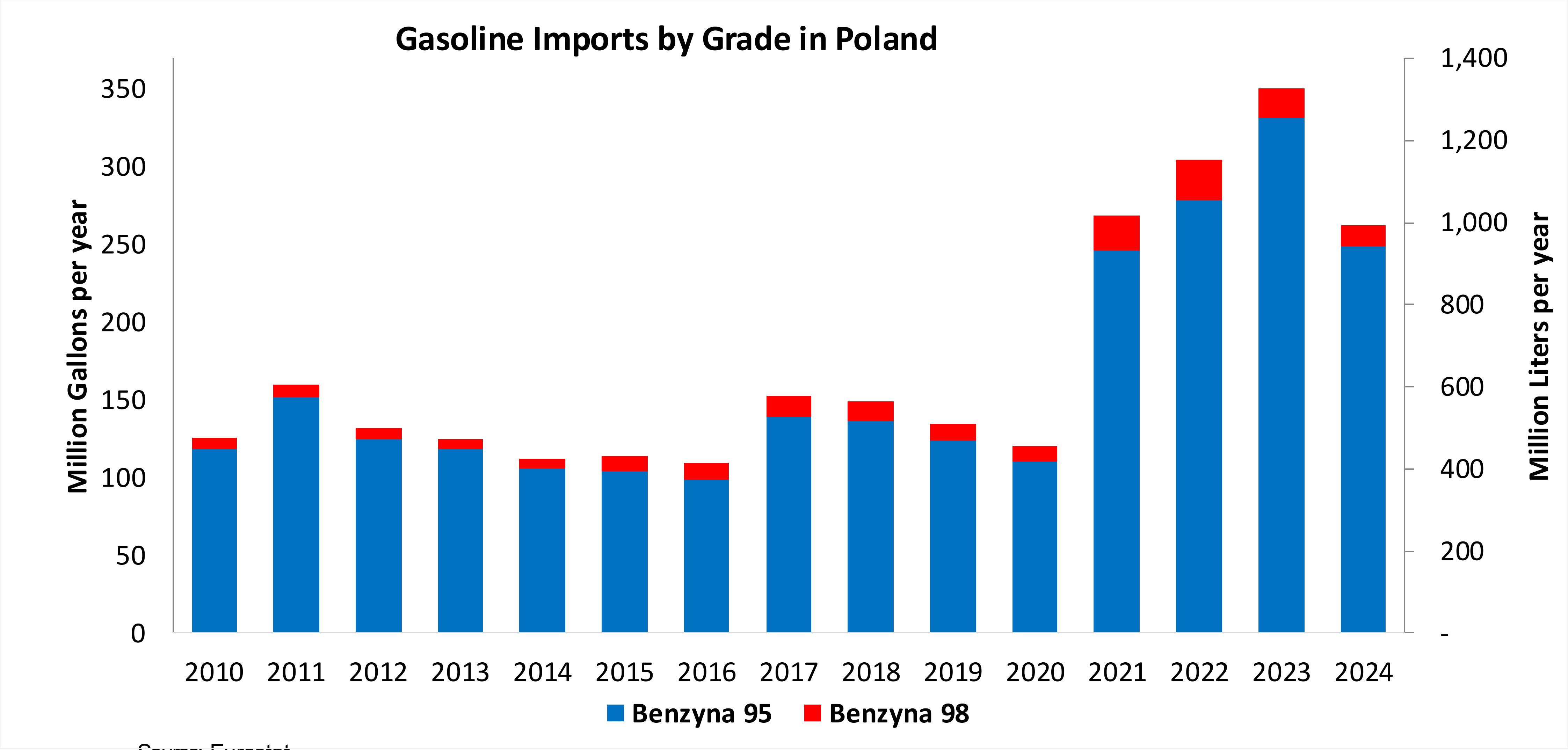
Source: Eurostat

Gasoline Production in Poland



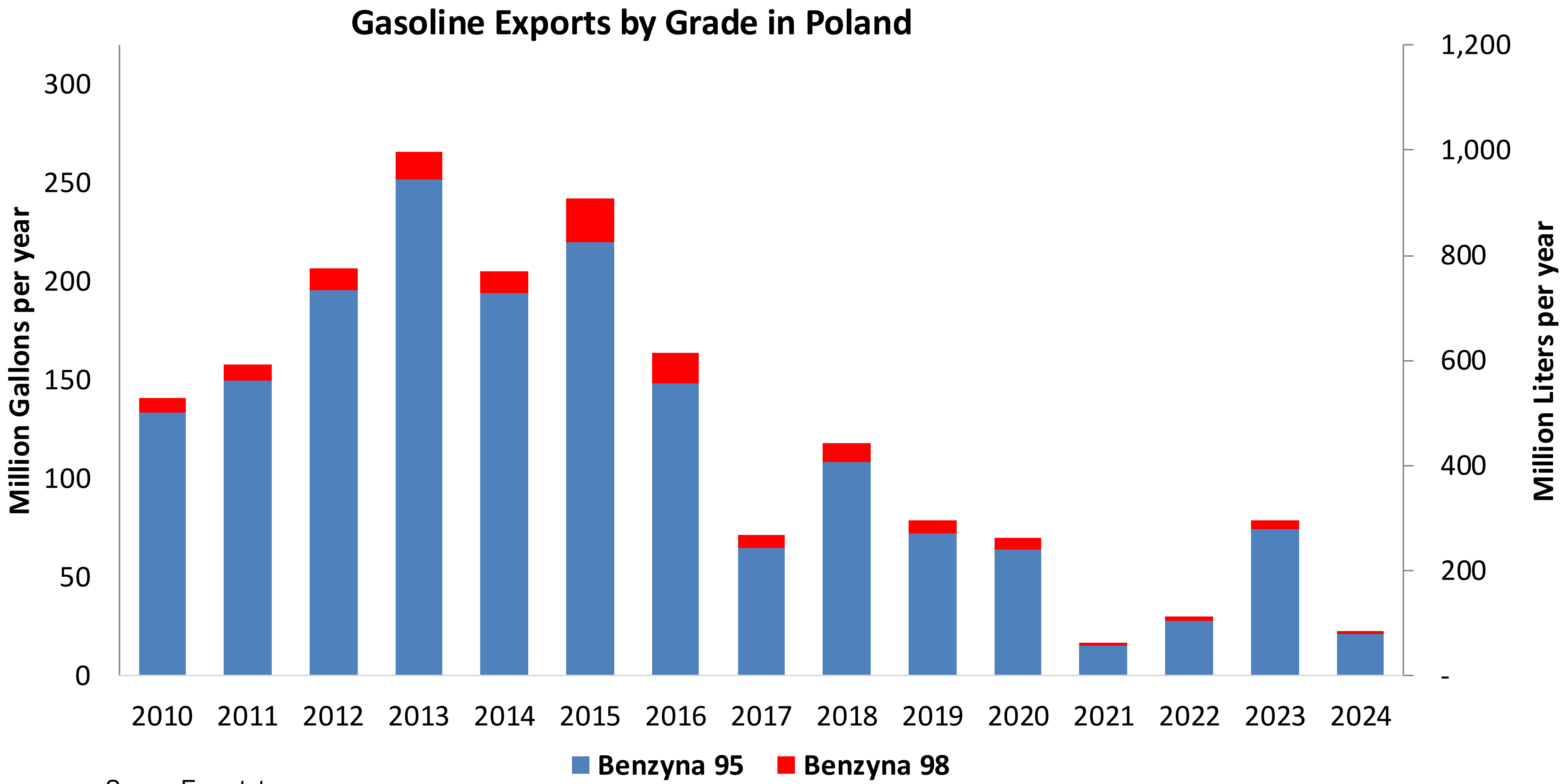
Source: Eurostat

Gasoline Imports in Poland



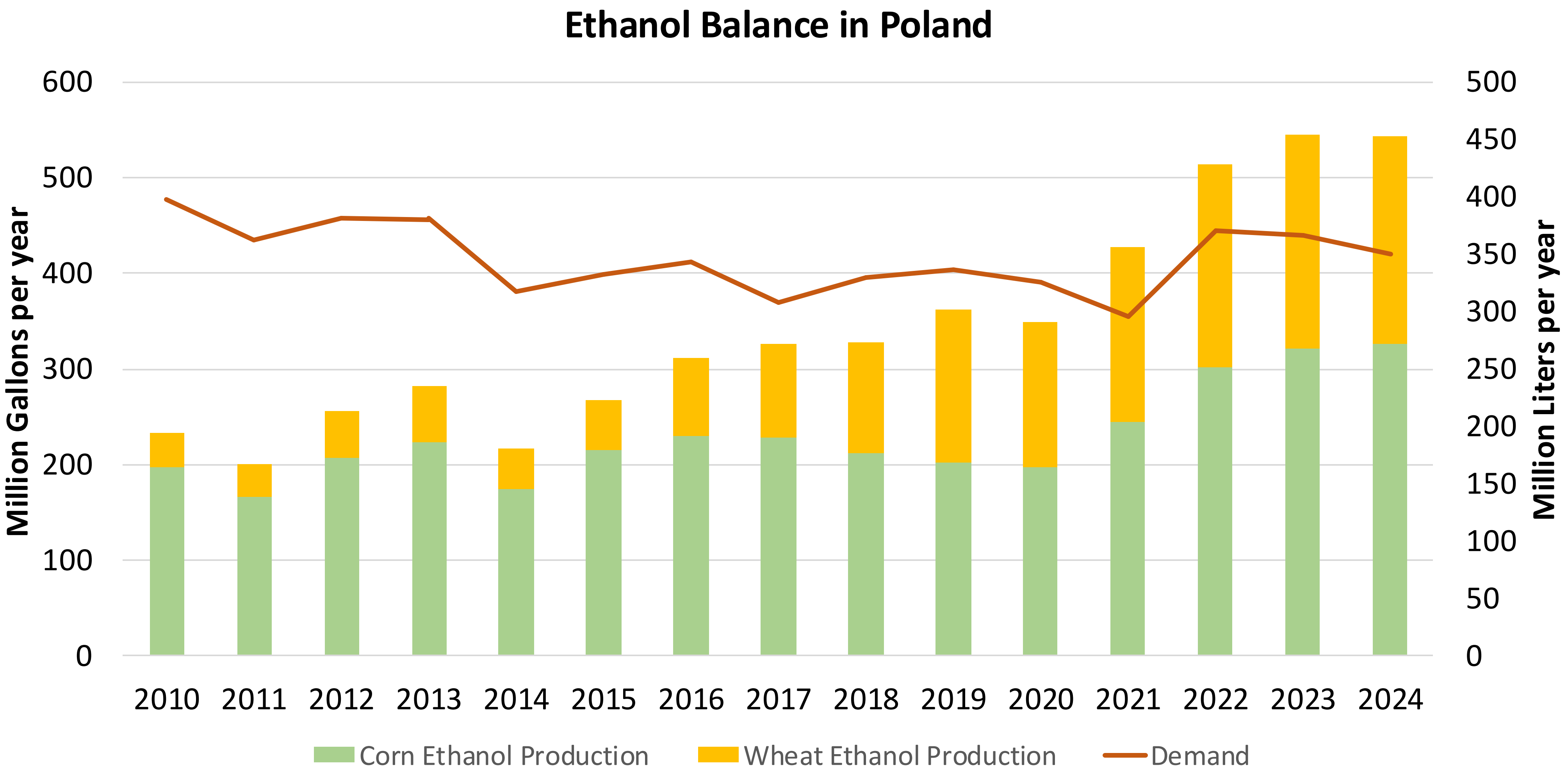
Source: Eurostat

Gasoline Exports in Poland



Source: Eurostat

Ethanol Balance in Poland



Source: Eurostat

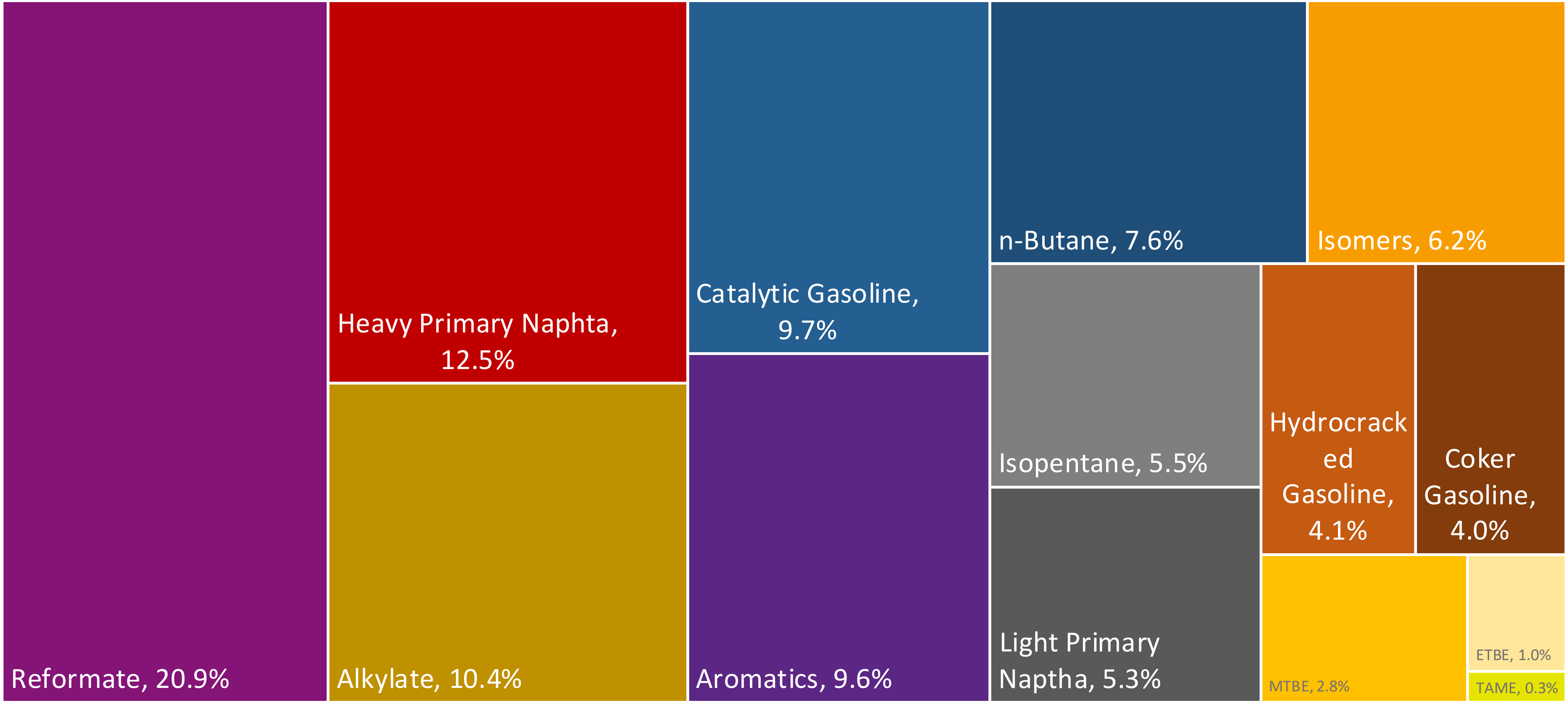
Gasoline Quality in Poland

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	9.2							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	4.6							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

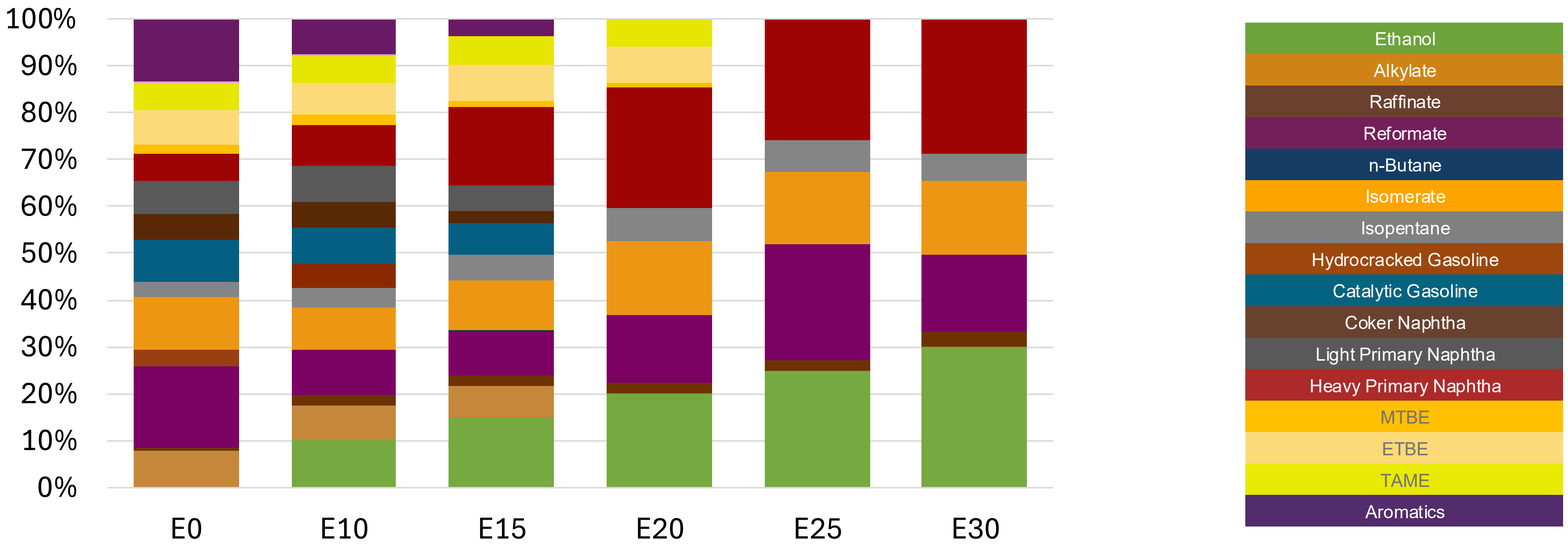
Source: European Commision

Poland Available Blending Components



Source: Faro90

Poland – Gasoline 95 – Constant Octane

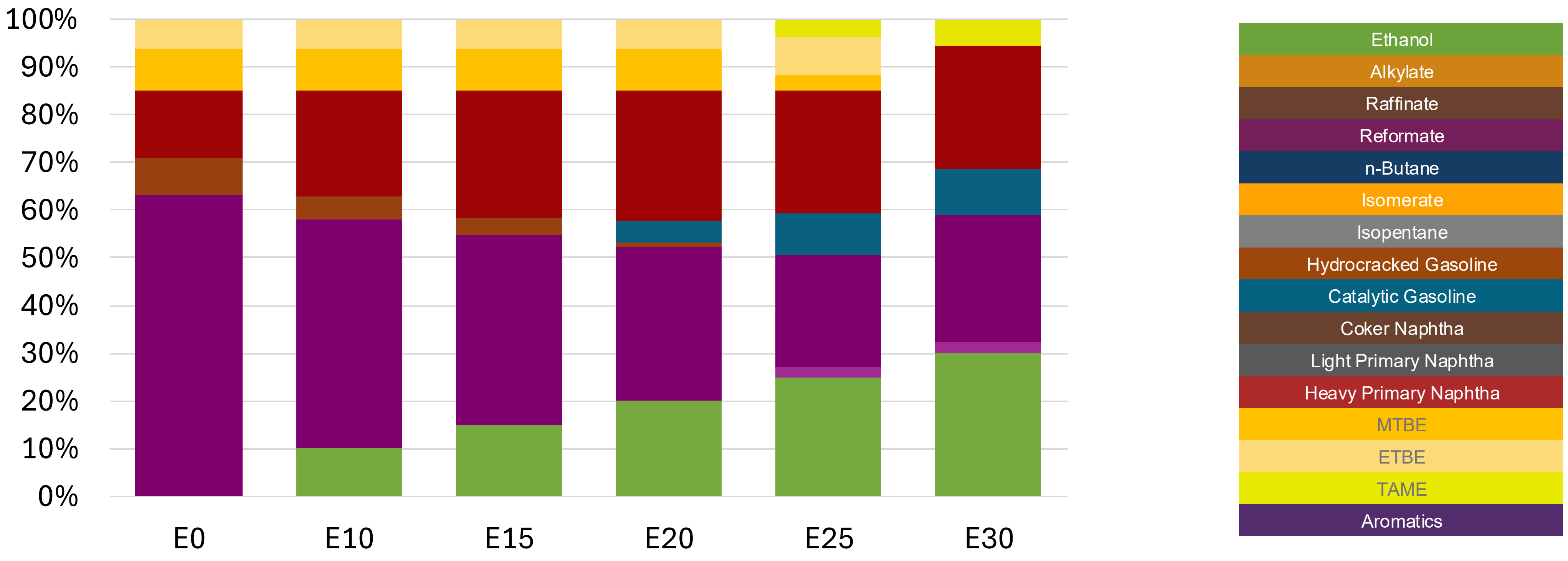


Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US$/gal)	$ 2.284	$ 2.146	$ 2.046	$ 1.931	$ 1.885	$ 1.832

Average CIF 2024 Prices

Source: Faro90

Poland - Gasoline 98 - Constant Octane

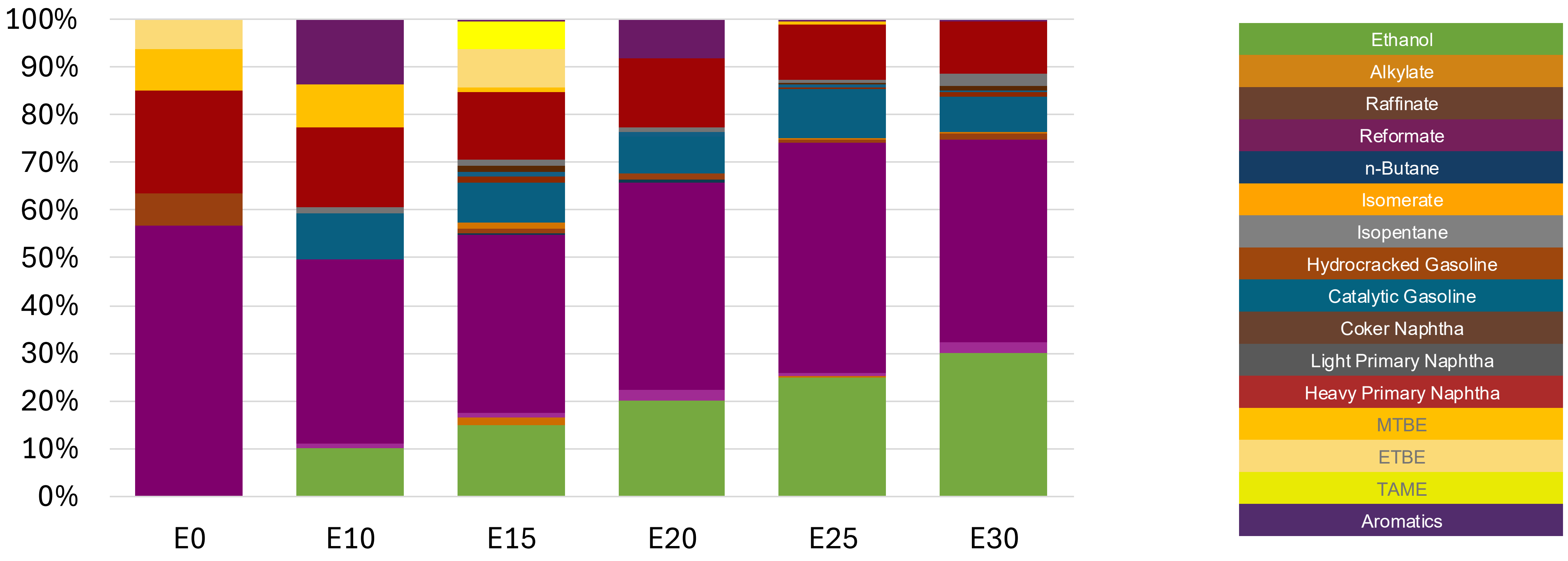


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.304	$ 2.187	$ 2.118	$ 2.046	$ 1.981	$ 1.931

Average CIF 2024 Prices

Source: Faro90

Poland – Gasoline 95 – Increased Octane

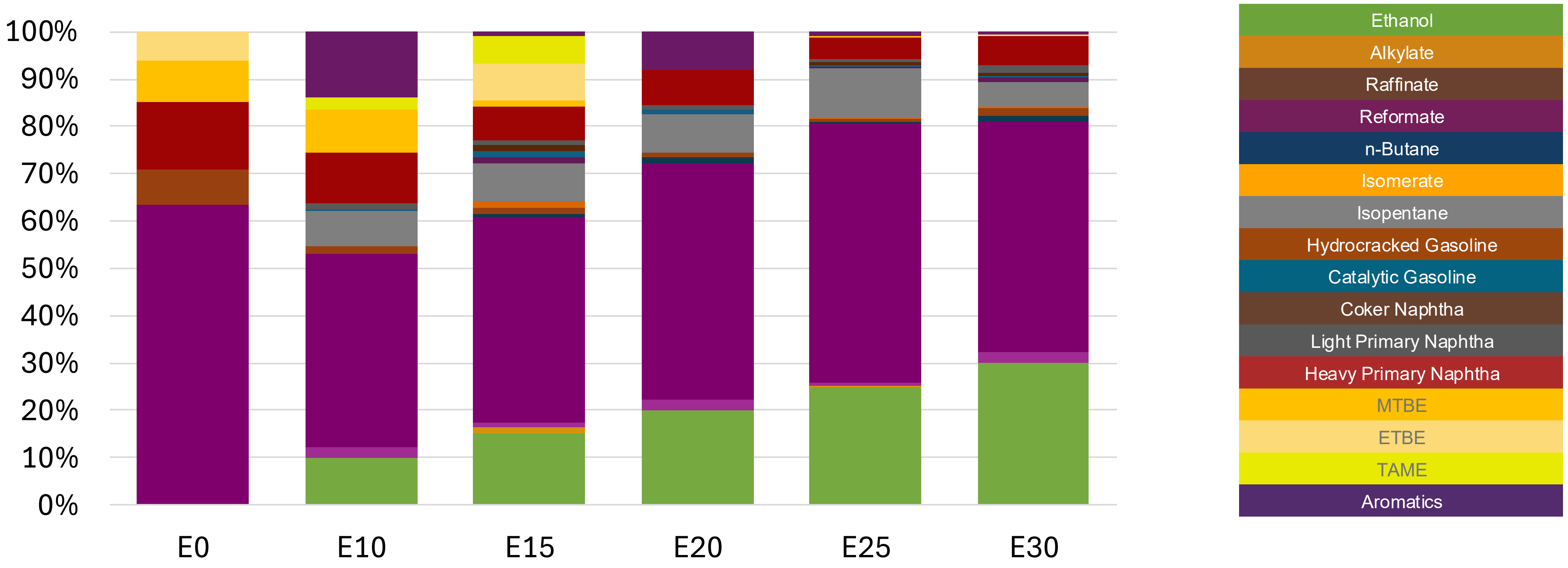


Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US$/gal)	$ 2.172	$ 2.172	$ 2.172	$ 2.172	$ 2.172	$ 2.172

Average CIF 2024 Prices

Source: Faro90

Poland – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US$/gal)	$ 2.304	$ 2.304	$ 2.304	$ 2.304	$ 2.304	$ 2.304

Average CIF 2024 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

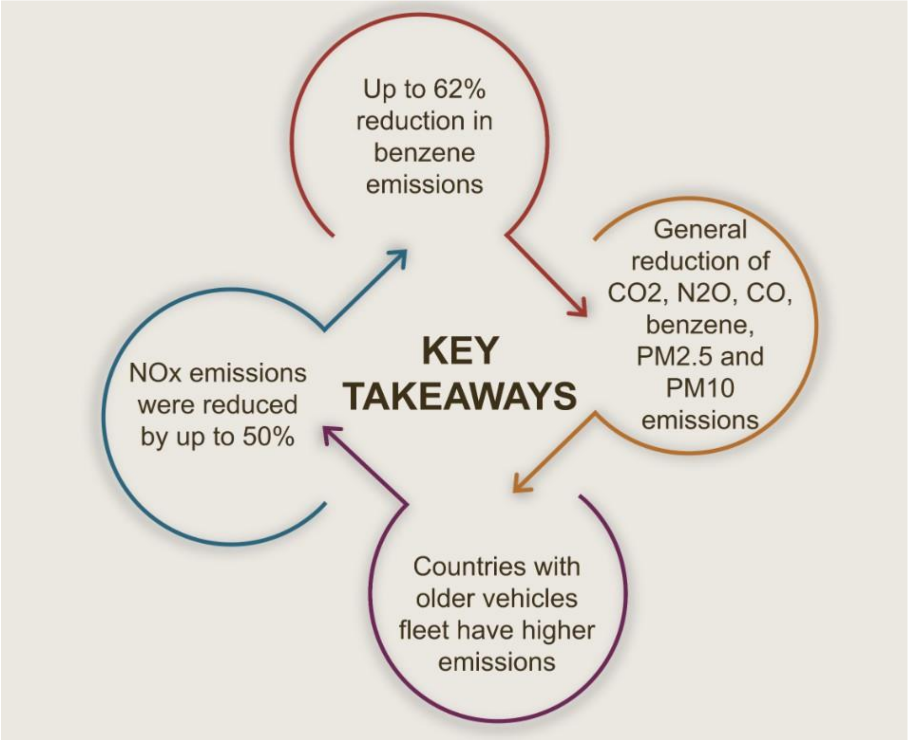
The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

Main Results



**Source: Bureau of transportation statistics.*

Gasoline Vehicle Fleet - Poland

Type	Weight	Motor	Escape	Recover	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	3,319
		Single FI	3-Ways		>8 years	132,626
		Multi FI	3-Ways		>8 years	1,203,420
			Euro III		>8 years	1,923,787
			Euro IV		>8 years	4,880,716
			Euro V		>8 years	1,755,479
			Euro 6		>8 years	3,588,663
					4-8 years	3,808,572
					<4 years	2,843,596
			Hybrid		All	86,307
		Electric			All	22,574
	Medium	Carburator	3-Ways	PCV Tank	>8 years	328
		Single FI	3-Ways		>8 years	3,924
		Multi FI	3-Ways		>8 years	167,390
			Euro III		>8 years	366,687
			Euro IV		>8 years	1,730,672
			Euro V		>8 years	537,754
			Euro 6		>8 years	1,196,221
					4-8 years	1,731,083
					<4 years	2,278,749
			Hybrid		All	67,747
		Electric			All	18,329
Sml Engines	Light	4-Cycle	None	PCV	>8 years	22,887
			Euro III		>8 years	23,628
			Euro IV		>8 years	360,111
			Euro V		>8 years	111,612
			Euro 6		>8 years	323,042
					4-8 years	447,003
					<4 years	973,260
			Hybrid		All	12,375
		Electric			All	3,286
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	16,574
		Fuel-Injection	3-Ways		>8 years	187,313
			Euro III		>8 years	105,052
			Euro IV		>8 years	121,096
			Euro V		>8 years	33,899
			Euro 6		>8 years	46,800
					4-8 years	22,064
					<4 years	2,867
			Hybrid		All	732
		Electric			All	2,005

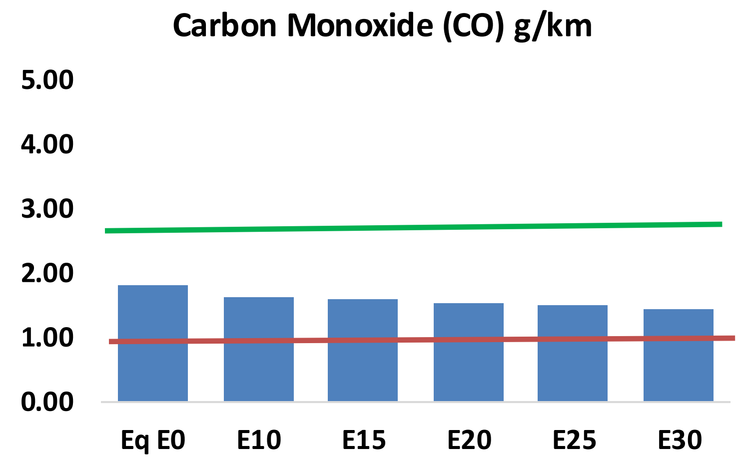
0 1,000,000 2,000,000 3,000,000 4,000,000 5,000,000 6,000,000

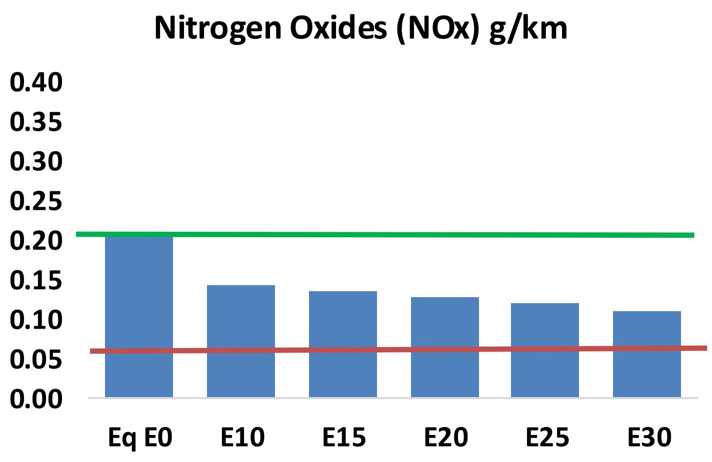
Vehicle Fleet: 31,163,548
Gasoline Vehicle Fleet: 13,217,757

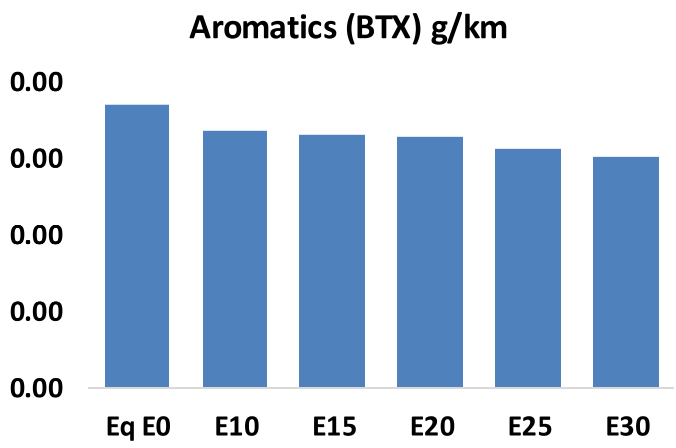
Source: Eurostat, ACEA

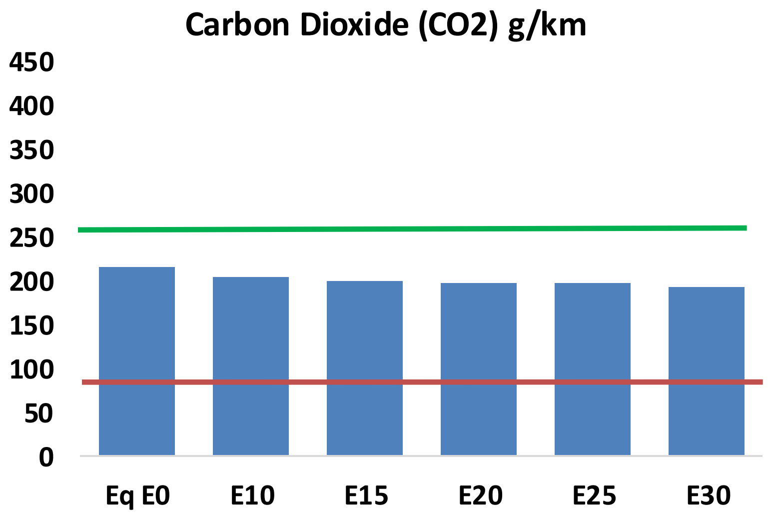
Poland – Vehicular Emissions

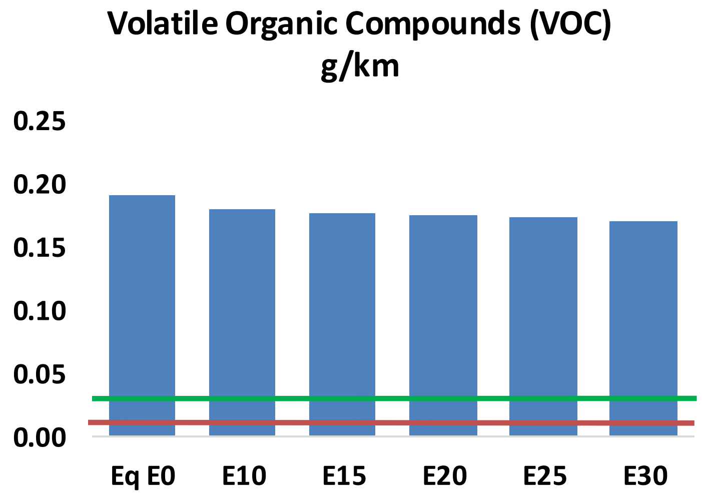
TIER III BIN 125 United States
Euro 6 Standard

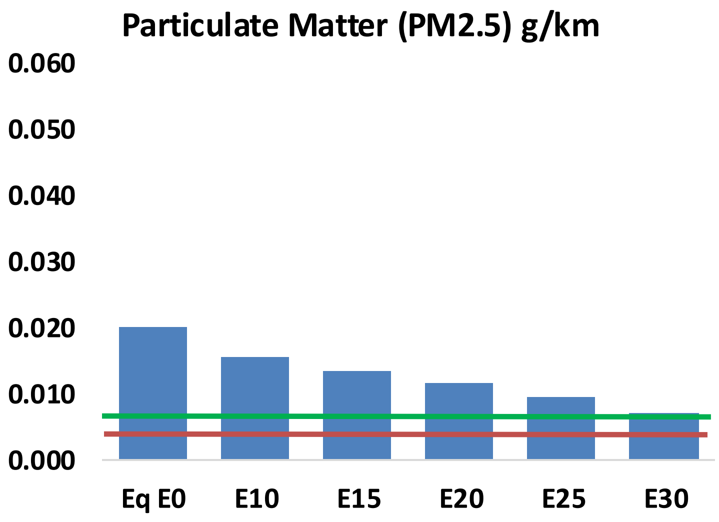












Source: Faro90

Poland – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	615,660	588,117	560,575	541,866	524,816	512,241	495,401	-9%	-15%
CO2	73,575,358	71,683,421	69,896,590	68,491,301	67,795,909	67,393,631	66,151,372	-5%	-8%
VOC	65,622	63,617	61,613	60,564	59,781	59,261	58,277	-6%	-9%
NOx	70,624	52,968	49,437	46,594	43,942	41,019	37,925	-30%	-38%
SOx	880	813	747	691	637	578	517	-15%	-28%
NH3	24,677	24,432	24,188	24,303	24,576	24,769	24,928	-2%	0%
N2O	1,147	1,140	1,136	1,141	1,165	1,178	1,191	-1%	2%
CH4	15,774	15,774	15,774	16,011	16,358	16,658	16,942	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	674	662	650	648	650	652	651	-4%	-4%
Acetaldehyde	1,351	1,550	2,275	3,464	4,720	5,393	6,373	68%	249%
Formaldehyde	4,074	3,961	4,617	5,421	5,680	6,283	6,840	13%	39%
Benzene	1,265	1,216	1,152	1,134	1,124	1,075	1,040	-9%	-11%
PM2.5	4,537	4,028	3,520	3,057	2,596	2,106	1,596	-22%	-43%
PM10	2,330	2,069	1,808	1,570	1,333	1,082	820	-22%	-43%

Source: Faro90

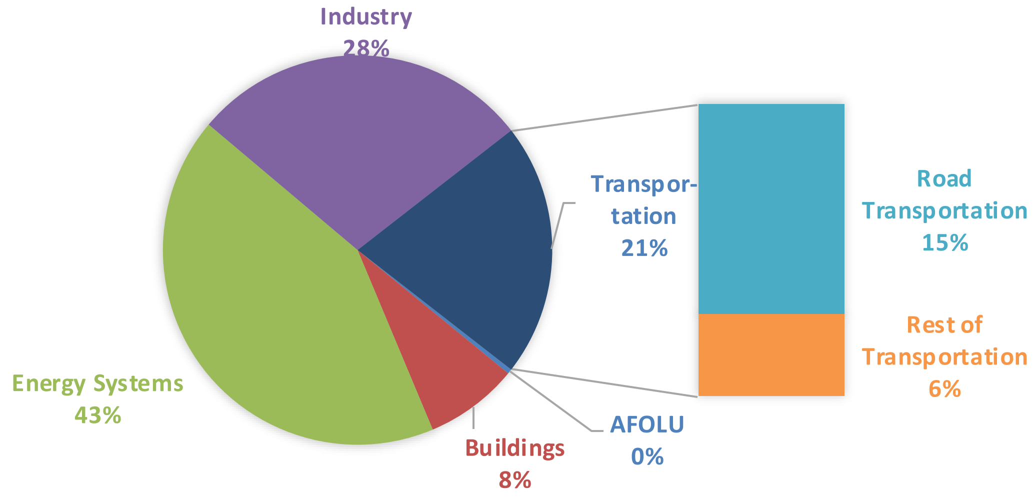
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

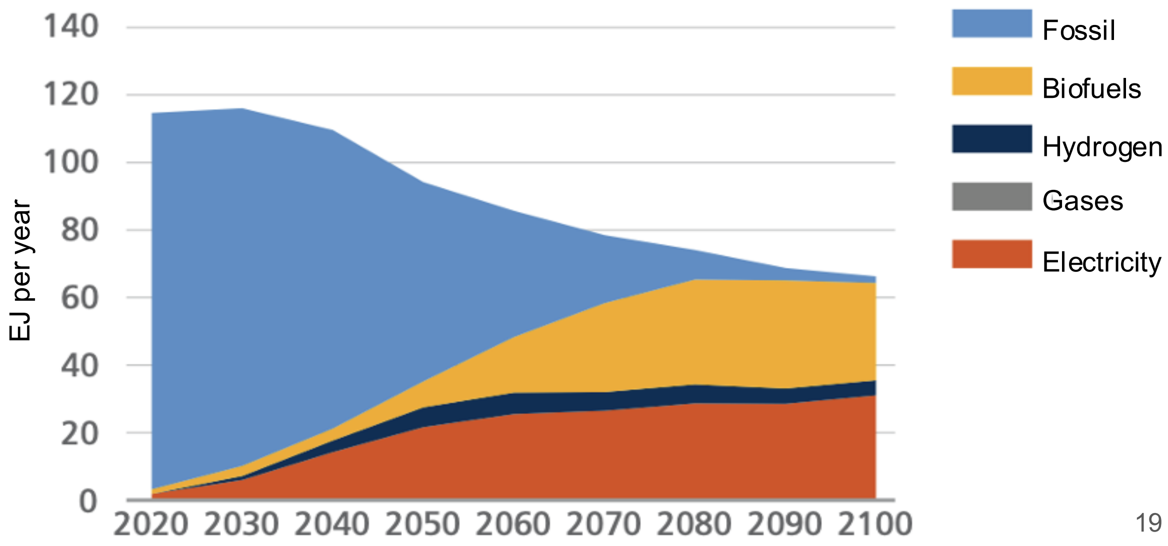
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

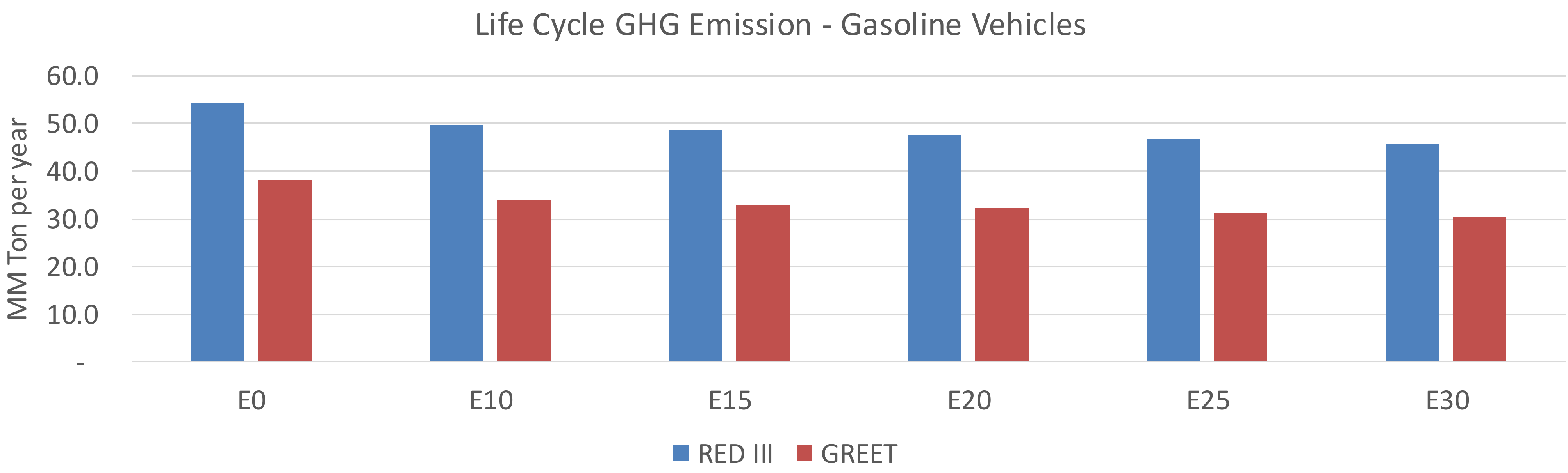
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Poland – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2024 (MMT/año)	337.8
Target @2035 (MMT/year)	190.2
Delta	147.7

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	10.9%	13.8%	16.0%	18.0%	20.9%
RED II/III	0.0%	8.1%	10.3%	12.0%	13.6%	15.7%

% Target@2035	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.8%	3.6%	4.1%	4.7%	5.4%
RED II/III	0.0%	3.0%	3.8%	4.4%	5.0%	5.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90